

Trust the Query, Not the Answer

Auditable Text-to-SPARQL over the GLEIF Register

Jannic Alexander Cutura | European Central Bank | DSTI School of Engineering

github.com/JannicCutura/finnlp-2026

1. The problem

In anti-money-laundering screening, credit-risk assessment and banking supervision an answer is useful only if it can be traced back to a source record. A language model asked directly does not meet that bar: ask a general-purpose chat assistant which entities Siemens consolidates and it returns about a dozen plausible names, none of them carrying the identifier you would need to check it. Ask the register and you get exactly **452** entities that report Siemens Aktiengesellschaft as their direct consolidation parent, each one traceable to its record by LEI.

So do not trust the answer. Trust the query, and read it.

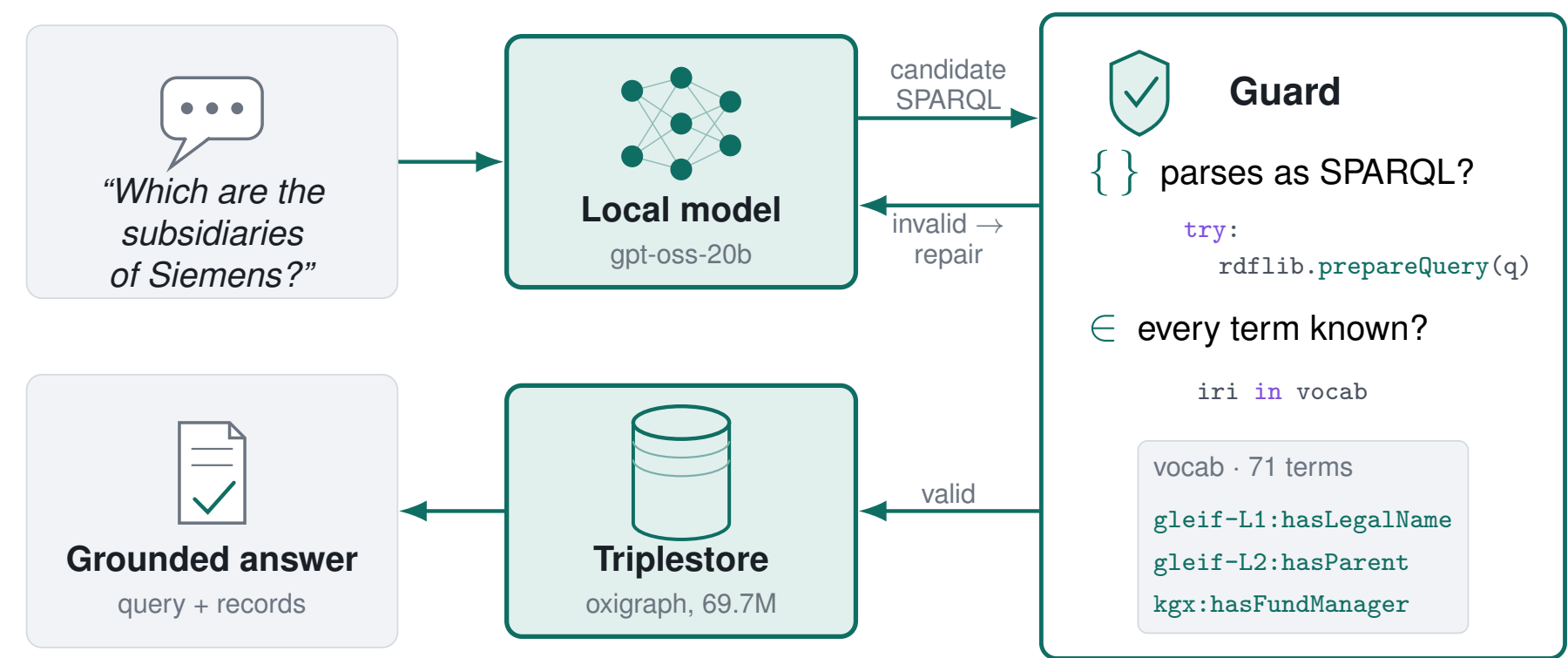
2. The register

GLEIF publishes the Legal Entity Identifier register as CC0 open data with an official RDF ontology — a real regulatory source rather than an encyclopaedic or synthetic graph. We pin one 8-hourly publish and query it locally.

Level 1 entity records	3,382,301
of which ISSUED	1,940,941
Level 2 relationships	481,557
Triples in the store	69.7 M
Populated predicates	27
Guard vocabulary	71 terms
Snapshot	20260723-1600
The 71-term guard vocabulary is everything a query may name: all 27 populated predicates, plus the 44 classes and enumerated code values they take. Every term is count-backed — it is in the prompt because it is in the graph.	

3. The system

A local open-weight model (gpt-oss-20b, served through LM Studio at temperature 0, prompting only, no training) turns the question into candidate SPARQL. A guard checks the query *before* it runs; a bounded repair loop of at most two retries feeds the guard's diagnostic back to the model. The query, its rows and the LEIs behind them are returned with every answer.



One shot, not agentic search: the guard, and not iterative graph exploration, is what keeps the output valid. The loop never retries on an empty result — for a question the register cannot answer, empty is the right answer.

4. Cost

Queries return in well under a second, except full-population aggregates at 14–28 seconds. Latency is dominated by generation (15–40 seconds), not by the store. The whole system runs on one machine and makes no external call, so the data never leaves the premises.

5. One question, end to end

A held-out item (h55), exactly as the system logged it. This is the whole audit trail an analyst sees.

Question business language, names no predicate

What is the legal form name of the direct consolidation parent of 'AMERICAN AXLE & MANUFACTURING, INC.'?

Attempt 1 model output — rejected by the guard

```
SELECT ?parentFormLabel WHERE {
  ?child a gleif-L1:RegisteredEntity ;
    gleif-L1:hasLegalName
      "AMERICAN AXLE & MANUFACTURING, INC." .
  ?child kgx:hasDirectConsolidationParent ?parent .
  ?parent gleif-base:hasLegalForm ?form .
  ?form rdfs:label ?parentFormLabel .
}
```

Guard: unknown vocabulary gleif-base:hasLegalForm — the predicate does not exist in the graph. Closest match: gleif-L1:hasLegalForm. The query never reaches the store.

Attempt 2 repaired — passes the guard, executes

```
PREFIX rdfs: <http://www.w3.org/2000/01/rdf-schema#>
SELECT ?parentFormLabel WHERE {
  ?child a gleif-L1:RegisteredEntity ;
    gleif-L1:hasLegalName
      "AMERICAN AXLE & MANUFACTURING, INC." .
  ?child kgx:hasDirectConsolidationParent ?parent .
  ?parent gleif-L1:hasLegalForm ?form .
  ?form rdfs:label ?parentFormLabel .
}
```

Answer, with the records behind it

```
parentFormLabel
"Corporation"

child lei:RY5TAKFOBLDUGX31MS24
parent lei:254900SMQY98VFKNCC32
form elf:XTIQ (ISO 20275)
```

6. What the guard does not do

The guard is an existence check, not a semantic one. It guarantees that the query parses and that every term it names is in the graph, which is what stops a hallucinated predicate from ever reaching the store. It does **not** guarantee that the query means what the question asked.

That boundary is the design and not a gap in it: a wrong answer arrives as a readable query returning wrong rows, never as fluent prose.

7. Where the errors are not

All twelve held-out errors are queries that ran and returned the wrong rows. None is invalid, none names a predicate that does not exist, and the model declines every out-of-scope and absent-entity question it is given.

Error type	n
String / entity matching	4
Literal typing	2
Wrong predicate / value / scope	3
Result shape / aggregation	3
Total	12

Every one of the twelve is visible in the query itself, which is the property the setting demanded.

8. Benchmark

144 questions over six tiers, written from a specification fixed *before* instantiation around a KYC analyst, a supervisor and a fund-operations user, in business language naming no predicate or IRI. Every reference query is executed against the pinned snapshot, so its result — the gold answer — cannot drift from the graph. The 54-question development set drove three schema revisions; the 90-question held-out set was written afterwards against a frozen prompt and informed none.

Tier	20B		120B
	Dev	Held	Held
a lookup	10/10	10/15	12/15
b single-hop	10/10	14/14	14/14
c multi-hop	8/8	13/15	14/15
d attribute filter	7/7	14/15	14/15
e aggregate	6/7	13/15	11/15
f unanswerable	12/12	14/16	14/16
Exec. accuracy	98.1%	86.7%	87.8%
95% CI	90–100	78–92	79–93
QALD macro-F1	0.981	0.897	0.893
Validity rate	100%	100%	98.9%
First-try / repaired	48/6	82/8	80/9

9. What each part buys

Held-out (20B)	Valid	Acc
Full system (guard + repair)	100%	86.7%
— guard & repair	91.1%	80.0%
— vocabulary	100%	32.2%
— schema	100%	13.3%

The in-prompt schema does most of the work: strip the vocabulary and accuracy falls to 32.2%, strip the schema and it collapses to 13.3%. The guard and the two-retry loop add the last 6.7 points and take validity to 100%.

10. Models

Held-out	Acc	95% CI	Valid
<i>Local, open-weight</i>			
gpt-oss-20b	86.7%	78–92	100%
gpt-oss-120b	87.8%	79–93	98.9%
Qwen2.5-Coder-32B	85.6%	77–91	97%
Llama-3.3-70B	70.0%	60–78	97%
<i>Non-local reference (not deployable)</i>			
gpt-5	90.0%	82–95	100%

Model size is not the bottleneck and open versus closed is not either; training focus is. The three code-tuned models sit inside one another's intervals, the general instruction model is 16 points below them, and the closed frontier reference leads by about three points.

11. Limits, and what is next

144 questions on one graph with a thin schema, one snapshot, one machine, one run per model, and no user study yet. The construction, however, is not specific to corporate structure: it is a model writing queries against any source described by an ontology. Real data sits across relational stores, graph stores, document stores and PDFs, each with its own query language — extending the same generate–guard–repair loop across them, routing each question to the store that can answer it and keeping every answer decomposable into a checkable query, is the direction we intend to pursue.

86.7% of 90 held-out questions answered correctly, every one with a valid query behind it — and every error visible in that query.

Code, prompts and benchmark
github.com/JannicCutura/finnlp-2026

The views expressed in this poster are those of the author and do not necessarily reflect those of the European Central Bank or the Eurosystem.